IGCSE 0607 Practice Questions — E10.4

Averages and measures of spread

Total Questions: 9 — **Total Marks:** 61 Marks

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1 The number of goals scored by a football team in each of their last 10 matches is shown below.

2 0 1 4 1 0 3 1 2 1

(a) Find the mode.

..... **[1]**

(b) Find the median.

..... **[2]**

(c) Calculate the mean.

..... **[2]**

2 A student records the number of points scored by a basketball team in 11 matches.

14 20 8 24 16 5 26 13 22 10 18

(a) Find the median.

..... **[1]**

(b) Find the range.

..... **[1]**

(c) Find the interquartile range.

..... **[2]**

(d) Calculate the mean.

..... **[2]**

3 The list shows the number of points scored by a basketball player in 7 games.

$$7 \quad 12 \quad 8 \quad 7 \quad 12 \quad x \quad 10$$

The mode of the number of points is 7.

(a) Write down the value of x.

$x =$ ..... **[1]**

(b) Find the range of the number of points.

..... **[1]**

(c) Find the median.

..... **[1]**

(d) Calculate the mean.

..... **[2]**

(e) In the 8th game, the player scores y points.
The new mean number of points for all 8 games is 9.5.
Find the value of y.

$y =$ ..... **[2]**

4 The table shows the heights, h cm, of 80 plants.

Height (h cm)	**Frequency**
$10 < h \leq 20$	12
$20 < h \leq 30$	25
$30 < h \leq 40$	28
$40 < h \leq 50$	11
$50 < h \leq 60$	4

(a) Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean height.

..... cm **[4]**

5 A survey was conducted on the masses, m grams, of 80 apples. The results are shown in the frequency table.

Mass (m grams)	$80 < m \leq 90$	$90 < m \leq 100$	$100 < m \leq 110$	$110 < m \leq 120$	$120 < m \leq 130$
Frequency	8	22	31	14	5

(a) Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean mass of the apples.

..... g **[4]**

(c) Explain why your answer to part (b) is an estimate.

.....

..... **[2]**

6 The table shows information about the time, t seconds, taken by 100 students to complete a puzzle.

Time (t seconds)	Frequency
$10 < t \leq 20$	14
$20 < t \leq 30$	28
$30 < t \leq 40$	32
$40 < t \leq 50$	18
$50 < t \leq 60$	8

(a) Write down the modal time interval.

..... **[1]**

(b) Calculate an estimate of the mean time.

..... seconds **[4]**

(c) Calculate the percentage of students who took strictly more than 40 seconds to complete the puzzle.

..... % **[2]**

(d) Write down the interval that contains the upper quartile.

..... **[1]**

7 A factory produces 60 items.
The masses, m kg, of these items are recorded in the table.
The estimate of the mean mass is 48 kg.

Mass (m kg)	**Frequency**
$0 < m \leq 20$	p
$20 < m \leq 40$	14
$40 < m \leq 60$	16
$60 < m \leq 80$	q
$80 < m \leq 100$	5

(a) Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[5]**

(b) State the class interval which contains the median mass.
You must show your working.

..... **[2]**

8 The mean of five numbers is 12.
The median of the five numbers is 11.
The mode of the five numbers is 8.
The range of the five numbers is 15.

(a) Find the five numbers.

..... **[5]**

(b) A sixth number, k, is added to the list.
The new mean is 13.
Work out the value of k.

$k =$ ..... **[2]**

9 The table shows information about the masses, m kg, of 50 parcels.

Mass (m kg)	Frequency
$0 < m \leq 2$	15
$2 < m \leq 4$	x
$4 < m \leq 6$	y
$6 < m \leq 8$	8
$8 < m \leq 10$	2

The estimate of the mean mass of the parcels is 3.8 kg.

(a) Use the total number of parcels to write down an equation in x and y, and show that it simplifies to $x + y = 25$.

[1]

(b) Use the estimate of the mean to show that $3x + 5y = 101$.

[4]

(c) Solve the simultaneous equations to find the value of x and the value of y.
You must show all your working.

$x =$ ..... , $y =$ ..... **[3]**

(d) Write down the interval that contains the median mass.

..... **[1]**